

Background Load (CPU)	Migration Duration	Time (w/o Mig)	Time (With Mig)
0	60	210s	259s
12	70	214s	255s
30	63	209s	261s

Fig. 3. Impact of Migration on application throughput with different background traffic

we conclude that it is possible to estimate the cost of live migration of each application for use by *pMapper*.

3.3 Power Modeling

The power-aware application placement controller explores various candidate placements and needs to estimate the overall power drawn for each candidate placement while selecting a good placement. This estimation is especially required to solve the power-constraints and power-performance tradeoff formulations. We next study the feasibility of modeling the power for a server given a mix of applications running on it.

Give a set of N applications and M servers, we can potentially mix a large variety of applications on each server. Further, any optimization algorithm may seek to change the ratio of various applications on a server. Hence, creating a model for all mixes of all applications on all the servers is practically infeasible. Thus, we may be able to estimate the power drawn by a server only if it is independent of the applications running on it. We next conducted experiments to validate this assumption by creating power models for various applications on the two testbeds.

We found (Fig. 4(a)) that the power drawn by a server varies with the applications running on it. Hence, an algorithm that requires an exact power model

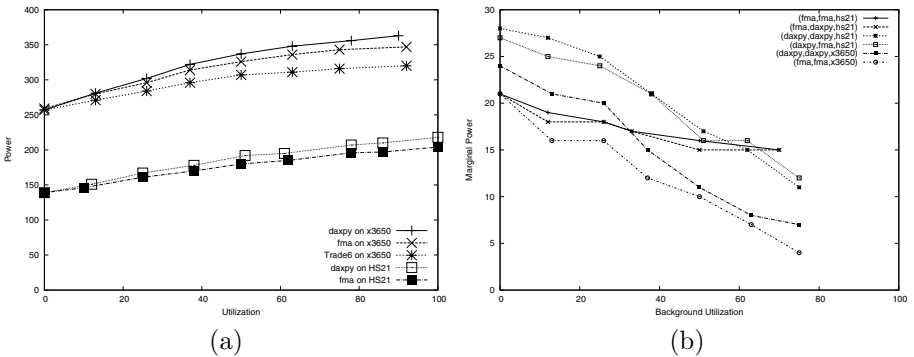


Fig. 4. (a) Power drawn by various applications with change in server utilization. (b) Marginal Power consumed by HPC applications on various platforms with change in background load and background mix. (A, B, C) denotes incremental power drawn by A with background traffic of B on platform C .